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Soft gluon resummation for associated squark-electroweakino production at the LHC [arXiv: 2202.13416 (hep-ph)]

J. Fiaschi, B. Fuks, M. Klasen, A. Neuwirth

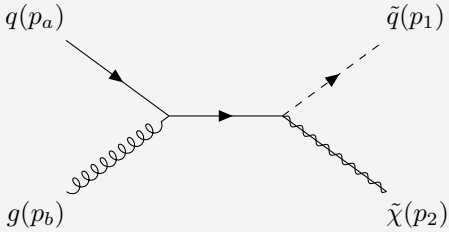
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Research Training Group

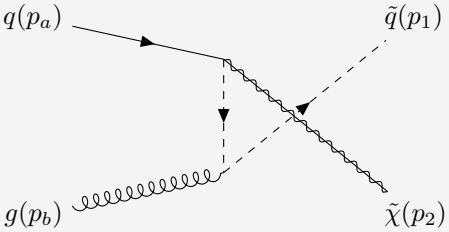
ICHEP 2022
BOLOGNA

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Squark and Electroweakino Production

- In the MSSM R-Parity implies production of SUSY particles in even numbers
- Expect dominant production of squark pairs and gluino pairs
- Production at the LHC might be restricted by heavy masses





- Squark and (lighter) electroweakino better LHC production candidate
- LO[1] and NLO[2] corrections are known

[1] Dawson, S. et al. Phys. Rev. D **31,** 1581 (1985).

[2] Binoth, T. et al. Phys. Rev. D **84,** 075005. arXiv: 1108.1250 [hep-ph] (2011), Frixione, S. et al. JHEP **12,** 008. arXiv: 1907.04898 [hep-ph] (2019), Baglio, J. et al. JHEP **12,** 020. arXiv: 2110.04211 [hep-ph] (2021).

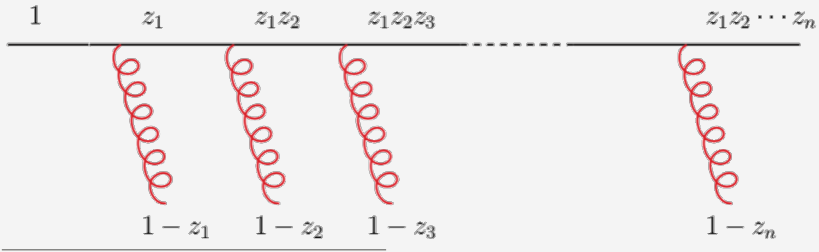
Resummation

- Massive particles are mostly produced close to threshold ($M^2/s = z \to 1$)[3]
- Perturbative expansion in α_s is spoiled by

Large Sudakov lograrithms ($m \leq 2n-1$)

$$\left(\frac{\alpha_s}{2\pi}\right)^n \frac{\log^m(1-z)}{1-z} \qquad \left(\frac{\alpha_s}{2\pi}\right)^n \log^m\left(\frac{M^2}{p_T^2}\right)$$

- Solution: Resummation of all soft emitted gluons



[3] Catani, S. & Trentadue, L. Nucl. Phys. B **327,** 323–352 (1989).

Resummation

- Transforming to Mellin space

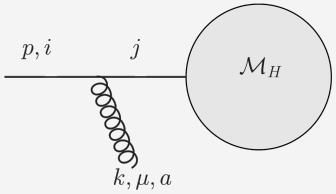
$$F(N) = \int_0^1 \mathrm{d}y\, y^{N-1} F(y) \quad \Longrightarrow \quad \left[\frac{\log^m(1-z)}{1-z}\right]_+ \overset{F}{\to} \sim \log^{m+1} N + \ldots$$

  factorizes the phase space

$$\int \mathrm{d}z z^{N-1} \mathrm{d}\phi_{2+n}(z) \approx \mathrm{d}\phi_2 \times \mathrm{d}\phi_n(N)$$

- Dynamical Factorisation is achieved by the eikonal approximation

$$\Gamma^{\mu}_{\text{eik}} = g_s T^j \frac{\Delta_i v^\mu}{\delta_i v \cdot k + i\epsilon}$$



Resummation

- From factorization theorem we split hard and long distance behaviour

$$\sigma(M^2, m^2) = \mathcal{H}(M^2/\mu_F^2)\mathcal{S}(m^2/\mu_F^2)$$

- Require independence on scale μ_F

$$\sigma(M^2, m^2) = \mathcal{H}(1)\mathcal{S}(1)\exp\left[-\int_{m^2}^{M^2}\frac{\mathrm{d}k}{k^2}\gamma_s(k^2)\right]$$

- From applying the RGEs we get[4]:

$$\sigma^{\text{Res.}}(N, M^2, \mu_R^2, \mu_F^2) = \mathcal{H}_{ab}(M^2, \mu_R^2, \mu_F^2)\exp\left[G_{ab}(N, M^2, \mu_R^2, \mu_F^2)\right] + \mathcal{O}\left(\frac{1}{N}\right)$$

with

$$G_{ab}(N, M^2, \mu_R^2, \mu_F^2) = \underbrace{LG_{ab}^{(1)}(N)}_{\text{Leading Log } \sim\log^2(N)} + \underbrace{G_{ab\to ij}^{(2)}(N, M^2, \mu_F^2, \mu_R^2)}_{\text{Next-to-Leading Log } \sim\log(N)} + \ldots$$

[4] Kidonakis, N. et al. Nucl. Phys. B **525,** 299–332. arXiv: hep-ph/9801268 (1998).

Matching

- Fixed order calculation with CS-dipole subtraction[5]:

$$\sigma^{\text{NLO}} = \int_3 \left[\mathrm{d}\sigma^{\text{R}} - \mathrm{d}\sigma^{\text{A}}\right]_{\epsilon=0} + \int_2 \left[\mathrm{d}\sigma^{\text{V}} + \int_1 \mathrm{d}\sigma^{\text{A}}\right]_{\epsilon=0} + \int_0^1 \mathrm{d}x \int_2 \left[\mathrm{d}\sigma^{\text{B}}(xp) \otimes (\mathbf{P}+\mathbf{K})(x)\right]_{\epsilon=0}$$

- Hard matching coefficients $C^{(i)}_{ab\to ij}(M^2, \mu_F^2, \mu_R^2)$
  - LO: $C^{(0)} = 1$

$$\mathcal{H}^{(0)}_{ab\to ij}(M^2, \mu_F^2, \mu_R^2) = \sigma^{(0)}_{ab\to ij}(M^2)$$

  - NLO: compute $C^{(1)} = \sigma^{(1)}_{ab\to ij}(M^2)/\sigma^{(0)}_{ab\to ij}(M^2)$ with suppressed 3 particle phase space contributions

$$\mathcal{H}^{(1)}_{ab\to ij}(M^2, \mu_F^2, \mu_R^2) = \sigma^{(0)}_{ab\to ij}(M^2)\, C^{(1)}_{ab\to ij}(M^2, \mu_F^2, \mu_R^2)$$

[5] Catani, S. et al. Nucl. Phys. B **627,** 189–265. arXiv: hep-ph/0201036 (2002).

Re-expansion

- To avoid double counting in:

$$\sigma_{ab} = \sigma_{ab}^{\text{NLO}} + \underbrace{\sigma_{ab}^{\text{Res.}} - \sigma_{ab}^{\text{Exp.}}}_{\sigma_{ab}^{\text{NLL}}}$$

- Subtract expansion in α_s:

$$\sigma_{ab}^{\text{Exp.}} = \mathcal{H}^{(0)}_{ab\to ij}(M^2,\mu^2) + \frac{\alpha_s}{2\pi}\mathcal{H}^{(1)}_{ab\to ij}(M^2,\mu^2) + \frac{\alpha_s}{2\pi}\mathcal{H}^{(0)}_{ab\to ij}(M^2,\mu^2)$$
$$\times \left(\left(A_a^{(1)} + A_b^{(1)} \right) \left(\log \bar{N} + \log \frac{\mu_F^2}{s} \right) - 2D^{(1)}_{ab\to ij} \right) \log \bar{N}$$

- PDFs fitted in Mellin space
- Inverse Mellin transformation with the integration contour following the principal value procedure and minimal prescription:

$$M^2 \frac{\mathrm{d}\sigma_{AB}}{\mathrm{d}M^2}(\tau) = \frac{1}{2\pi i} \int \mathrm{d}N \tau^{-N} M^2 \frac{\mathrm{d}\sigma_{AB}}{\mathrm{d}M^2}(N)\,.$$

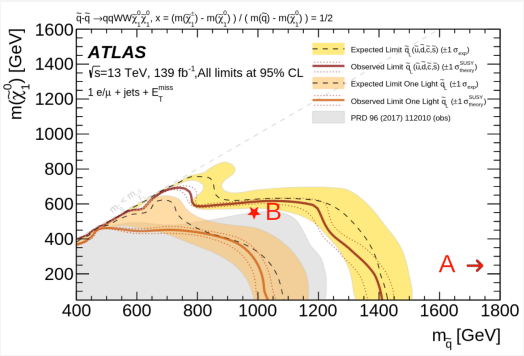
Scenarios

pMSSM-11 scenario A [TeV]		
M_1	M_2	M_3
0.25	0.25	-3.86
$M_{(U,D,Q)_{1,2}}$	$M_{(U,D,Q)_3}$	μ
4.0	1.7	1.33
$M_{(L,E)_{1,2}}$	$M_{(L,E)_3}$	$\tan\beta$
0.35	0.47	36
M_A	A_0	
4.0	2.8	
$m_{\tilde{\chi}_1^0}$	$m_{\tilde{u}}$	$m_{\tilde{g}}$
0.249	4.07	3.90

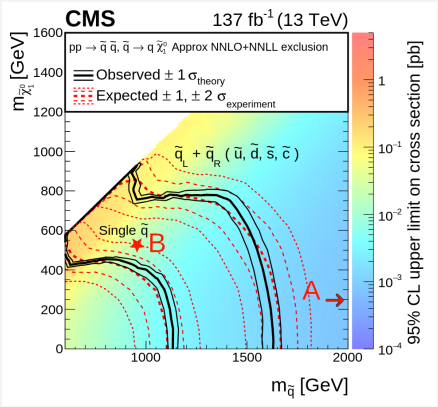
pMSSM-11 scenario B [TeV]		
M_1	M_2	M_3
0.51	0.48	3.00
$M_{(U,D,Q)_{1,2}}$	$M_{(U,D,Q)_3}$	μ
0.9	2.0	-9.4
$M_{(L,E)_{1,2}}$	$M_{(L,E)_3}$	$\tan\beta$
1.85	1.33	33
M_A	A_0	
3.0	-3.4	
$m_{\tilde{\chi}_1^0}$	$m_{\tilde{u}}$	$m_{\tilde{g}}$
0.505	0.96	2.94

Current Limits

ATLAS[6]



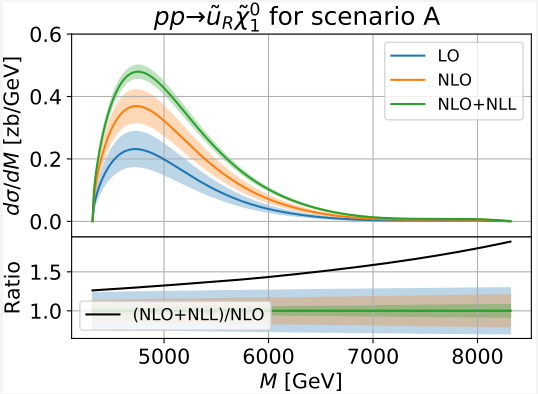
CMS[7]

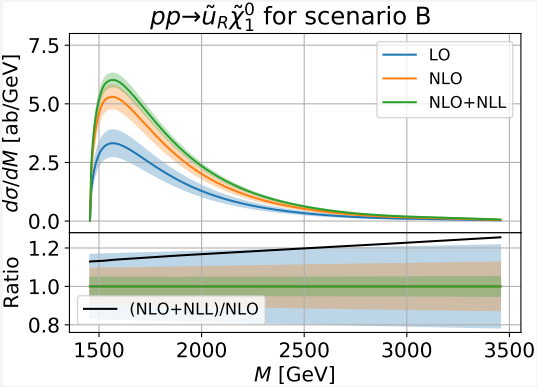


[6] Aad, G. et al. Eur. Phys. J. C **81.** [Erratum: Eur.Phys.J.C 81, 956 (2021)], 600. arXiv: 2101.01629 [hep-ex] (2021).
[7] Sirunyan, A. M. et al. JHEP **10,** 244. arXiv: 1908.04722 [hep-ex] (2019).

Invariant Mass

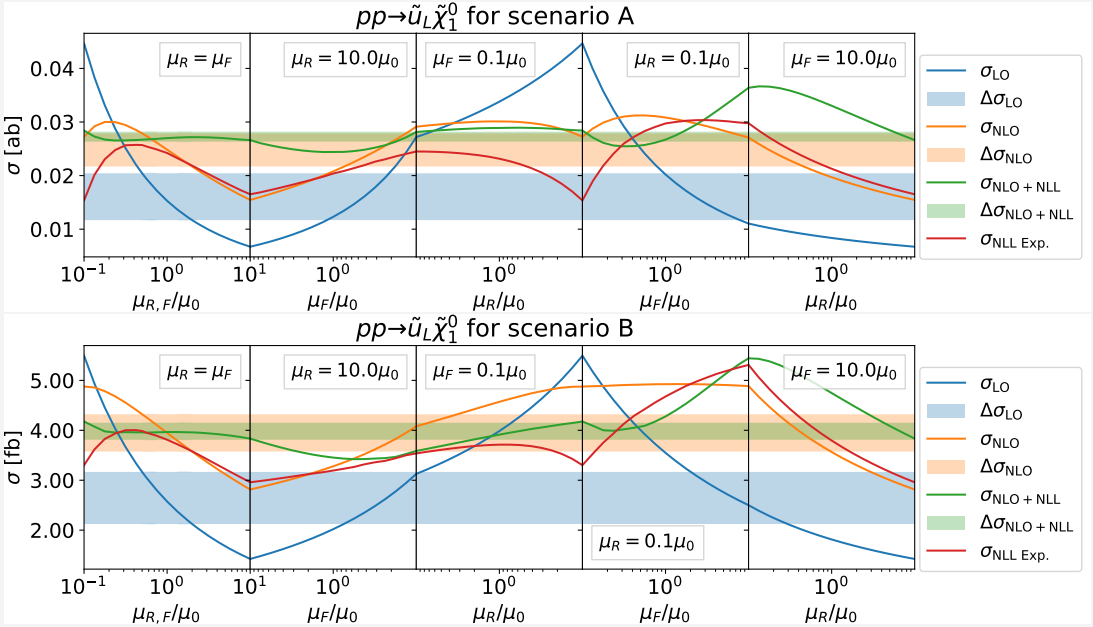
- $\sqrt{S} = 13$ TeV and MSHT20 PDF





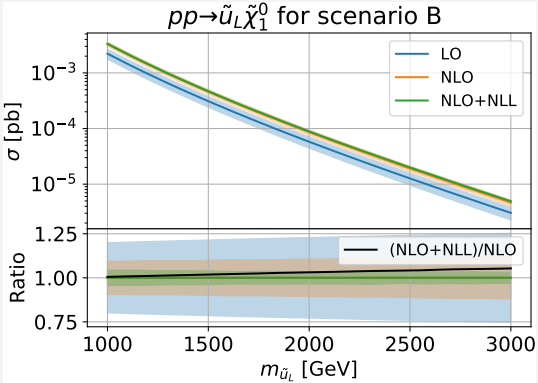
- Large resummation contributions very close to threshold
- Seven-point method around scale $\mu = M$ to estimate scale uncertainties

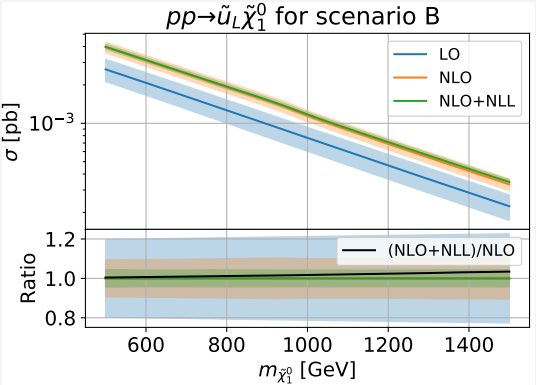
Scale Variation



Note: panels 2 and 3 are inverted.

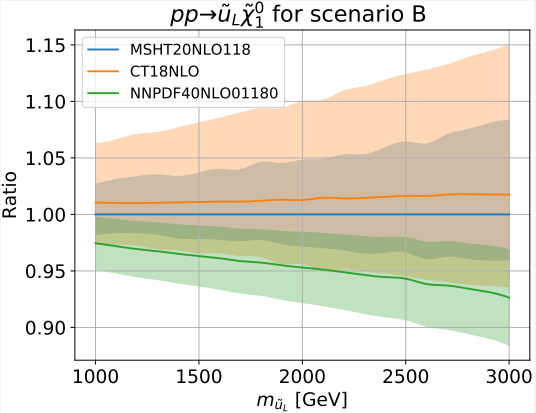
Total Cross sections

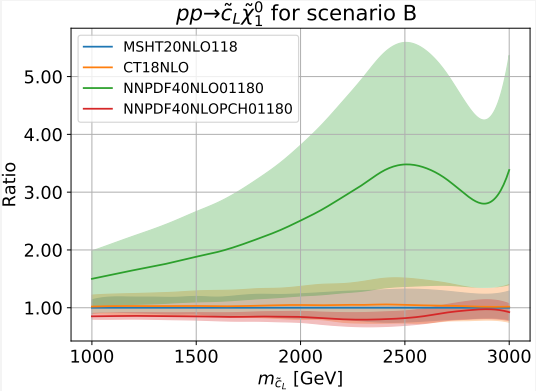




- NLL contributions of up to 6%
- Scale uncertainty reduced to ‹5%

Parton density uncertainties





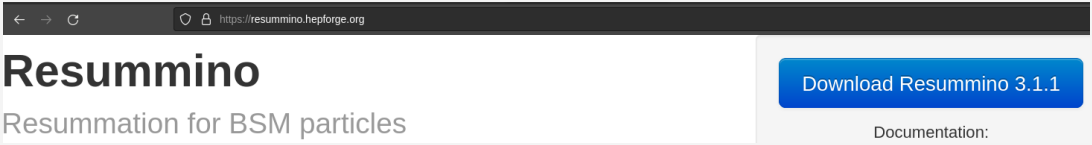
- Up-PDF uncertainties at 90% CL are consistent
- For 2 TeV up-squarks they range from $\pm 4\%$ to $\pm 7\%$
- Difference in NNPDF40's fitted charm and perturbative charm treatment.

Conclusion

- Production of squark/gluino with gaugino relevant if squark and gluino pair production are too heavy for the LHC
- We computed QCD-NLO corrections with CS-dipoles and OS-subtraction for the NLL matching of soft gluon emissions
- Resummation increases the σ_{tot} by 2% to 6% for squark masses ranging from 1 TeV to 3 TeV
- Renormalisation and Factorisation scale dependence is reduced from 20% (LO) to 10% (NLO) to <5% (NLO+NLL)
- Current PDF uncertainties are of the same order as scale uncertainties after Resummation

Resummino

- Resummino 3.1.1 code published on HEPForge:

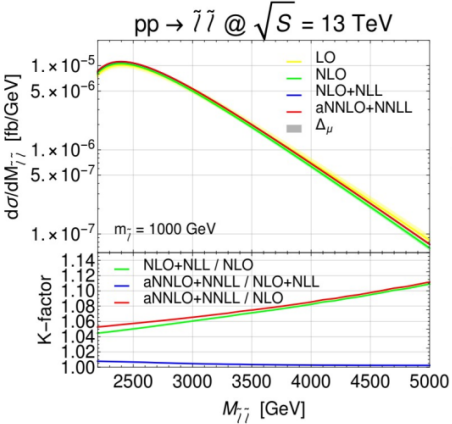


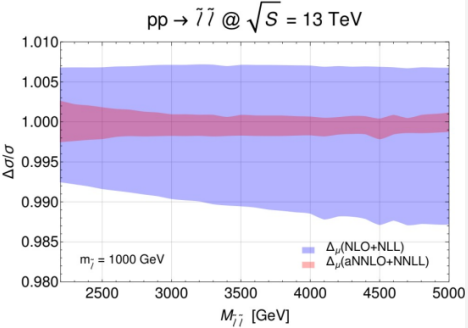
- approximate NNLO+NNLL for slepton-pair[8], electroweakino-pair[9], Z' and W'[10] production
- NLO+NLL for gaugino-gluino[11] and gaugino-squark[12] production

[8] Fiaschi, J. et al. JHEP **04,** 049. arXiv: 1911.02419 [hep-ph] (2020).
[9] Fiaschi, J. & Klasen, M. Phys. Rev. D **102,** 095021. arXiv: 2006.02294 [hep-ph] (2020).
[10] Jezo, T. et al. JHEP **12,** 092. arXiv: 1410.4692 [hep-ph] (2014).
[11] Fuks, B. et al. JHEP **07,** 053. arXiv: 1604.01023 [hep-ph] (2016).
[12] Fiaschi, J. et al. JHEP **06,** 130. arXiv: 2202.13416 [hep-ph] (2022).

Thank you!

Slepton pair production

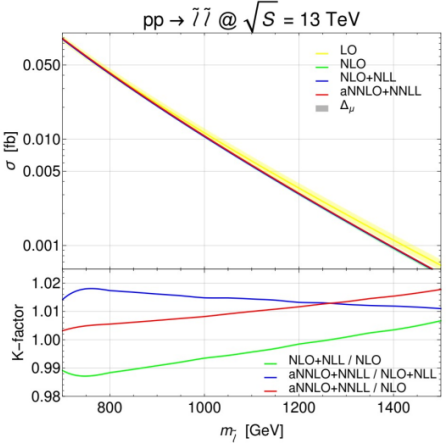


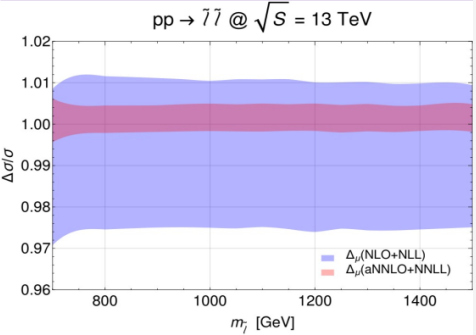


- Visible reduction of scale uncertainty from about 1% to about 0.1%.

- Resummation effects at NLL are more important at production threshold.
- K-factor increases from 4.5% to 11%
- The increase from NLO+NLL to aNNLO+NNLL is about 1%

Slepton pair production



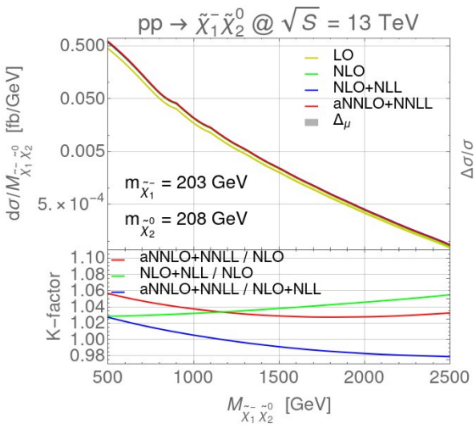


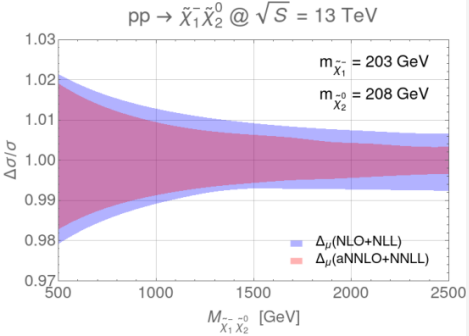
- Visible reduction of scale uncertainty from few % to few 0.1%.

- These cross sections correspond to:
  - more than 10 events at *700 GeV* at current integrated luminosity of *139 fb*$^{-1}$;
  - 3 events at *1 TeV* for LHC Run 3 goal of *300 fb*$^{-1}$;
  - Few events at *1.5 TeV* for HL-LHC goal of *3 ab*$^{-1}$

- aNNLO(+NNLL) corrections increase the NLO(+NLL) prediction by 1% - 2%.

Mostly Higgsino Electroweakinos

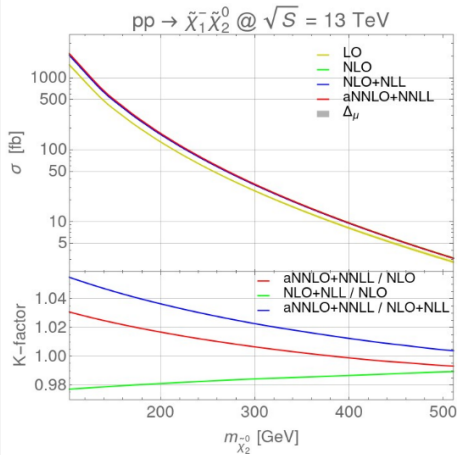


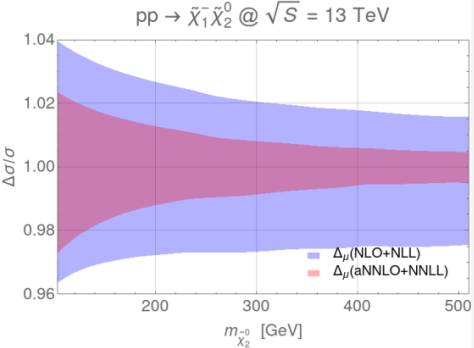


- Visible reduction of scale uncertainty from about from ±2.1% to ±1.8% (±0.6% to %0.4%) at low (large) invariant masses.

- NLO corrections enhance the LO cross section by about 30%.
- NLL corrections enhance the NLO cross section by another 3-5%.
- aNNLO+NNLL give another ±2 % correction to the NLO+NLL cross section.

Mostly Higgsino Electroweakinos



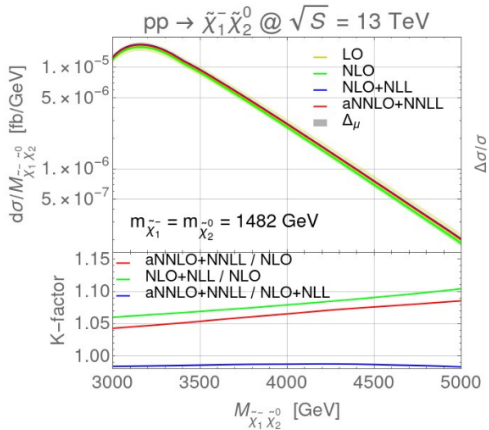


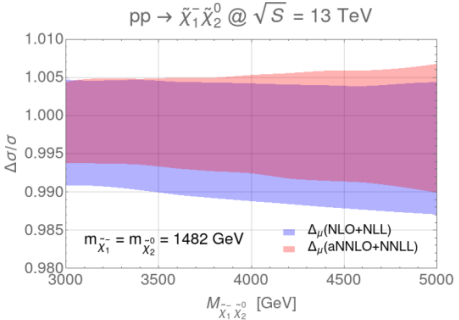
- Visible reduction of scale uncertainty from about from ±4% to ±2.5% (+1.5% and -2.5% to ±0.5%) for light (heavy) Higgsinos

(Scale uncertainties are not so small, because of the relatively light Higgsino masses)

- NLL corrections reduce the NLO cross section by 1% to 2%.
- aNNLO+NNLL corrections increase the NLO+NLL cross section by up to 5% for low higgsino masses.

Mostly gauginos Electroweakinos

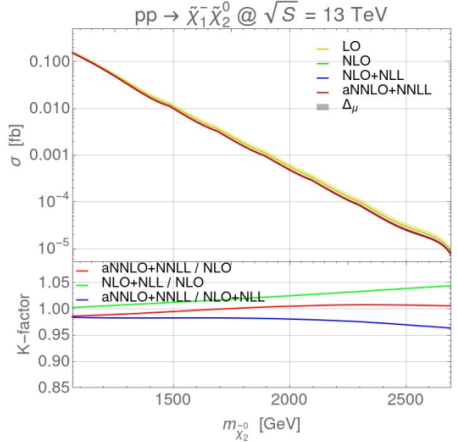


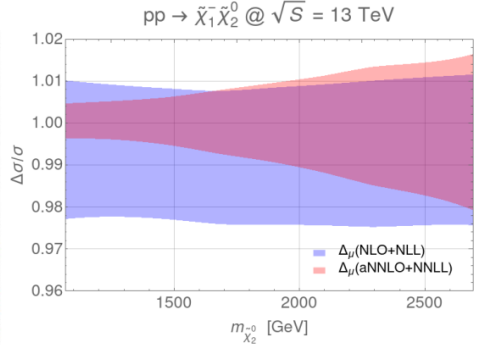


- Reduction of scale uncertainty on the lower bound from – 0.9% to –0.6% at low invariant masses.

- NLL corrections increase the cross section between 5% to 10%.
- A decrease for all invariant masses is observed from NLO+NLL to aNNLO+NNLL.
- This behavior is correlated with large *t*- and *u*- channel contributions and large cancellations of the squared *s*- channel contribution with its interference terms.

Mostly gauginos Electroweakinos

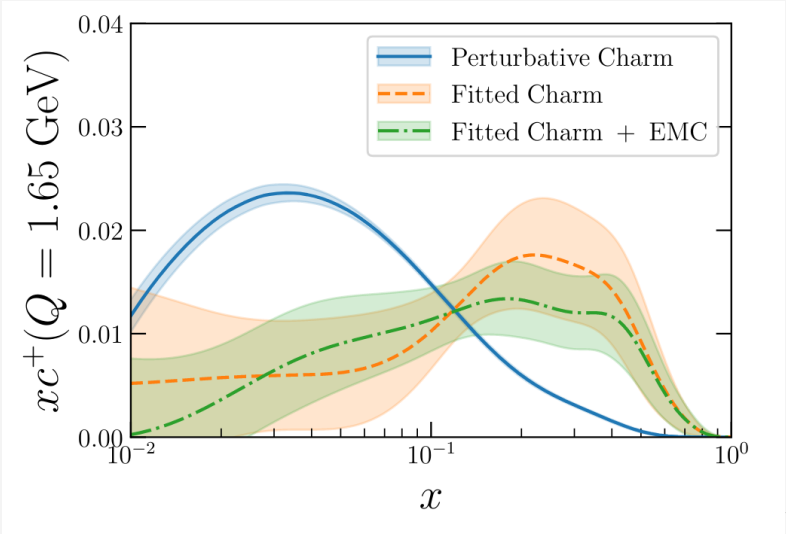




- Scale uncertainty gets reduced from 3% to about 1% for light gauginos, while it is stable at around 3% for heavy gauginos

- NLL corrections increase the NLO cross section by up to 5%.
- aNNLO+NNLL corrections reduce the NLO+NLL cross section by up to 4% for high gaugino masses.

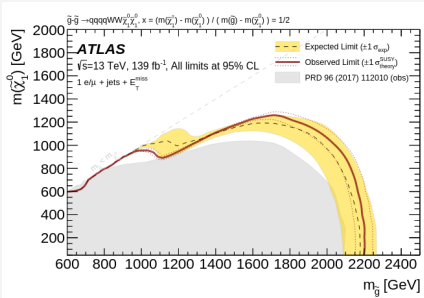
NNPDF40 charm

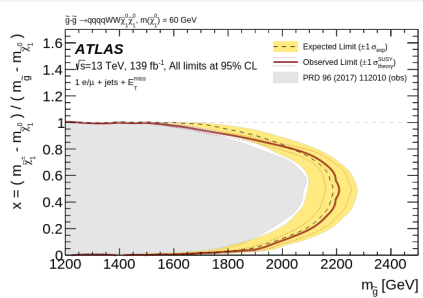


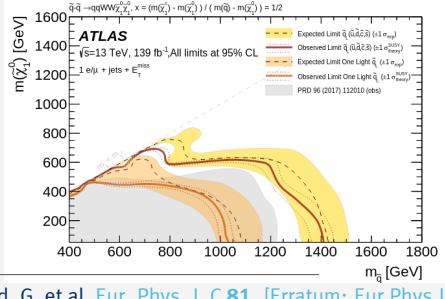
13

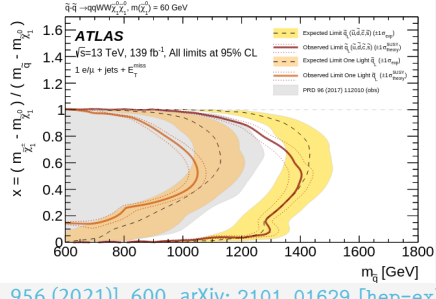
[13] Rojo, J. in 55th Rencontres de Moriond on QCD and High Energy Interactions (Apr. 2021). arXiv: 2104.09174 [hep-ph].

ATLAS



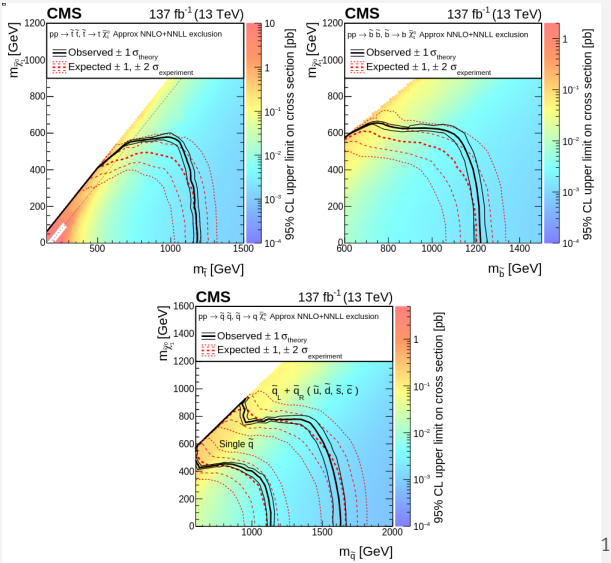






[14] Aad, G. et al. Eur. Phys. J. C **81.** [Erratum: Eur.Phys.J.C 81, 956 (2021)], 600. arXiv: 2101.01629 [hep-ex] (2021).

CMS



[15] Sirunyan, A. M. et al. JHEP **10,** 244. arXiv: 1908.04722 [hep-ex] (2019).

PDF fitting

To perform our computation in Mellin space we still need to transform the PDFs. This is achieved by fitting them in x-space to the function used by the MSTW collaboration

$$f(x) = A_0 x^{A_1}(1-x)^{A_2}(1 + A_3\sqrt{x} + A_4 x + A_5 x^{\frac{3}{2}}) + A_6 x^2 + A_7 x^{\frac{5}{2}} \,,$$

with the analytical result in Mellin space

$$\begin{aligned} F(x) = A_0\Gamma(y)B'(A_1+N,y) + A_3 B'(A_1+N+\frac{1}{2},y) + A_4 B'(A_1+N+1,y) \\ + A_5 B'(A_1+N+\frac{3}{2},y) + A_6 B'(A_1+N+2,y) + A_7 B'(A_1+N+\frac{5}{2},y) \end{aligned}$$

and

$$B'(x,y) = B'(x,A_2+1) = B(x,y)/\Gamma(y) = B(x,y)/\Gamma(y)\,.$$